

Data and scope

Public GitHub PR traces



GitHub GraphQL

Pull Requests + Comments + Commits + Issues + Reviews + User Contribution + metadata

Collected Dataset

Agent PRs

325,523

Human PRs

29,808

Bot PRs

1,206

Window

2025.04–2026.01

Public user profiles

Active contributors

56,190

Median tenure

5.35 years

IQR tenure

1.99–10.14

Seniority strata within profiled subset

Novice

25%

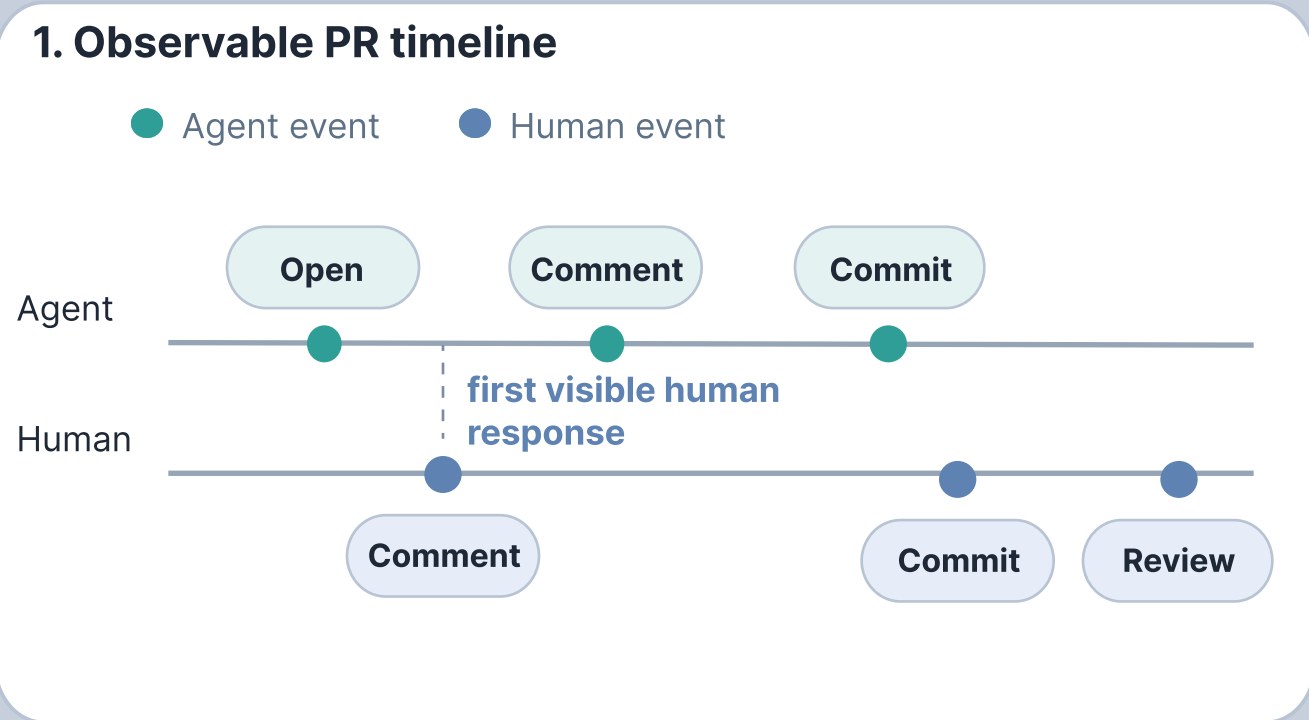
Mid

50%

Expert

25%

From traces to interactive pathways



Reconstructed PR Timelines (N = 205250)

With human involvement

78.4%

With human commits

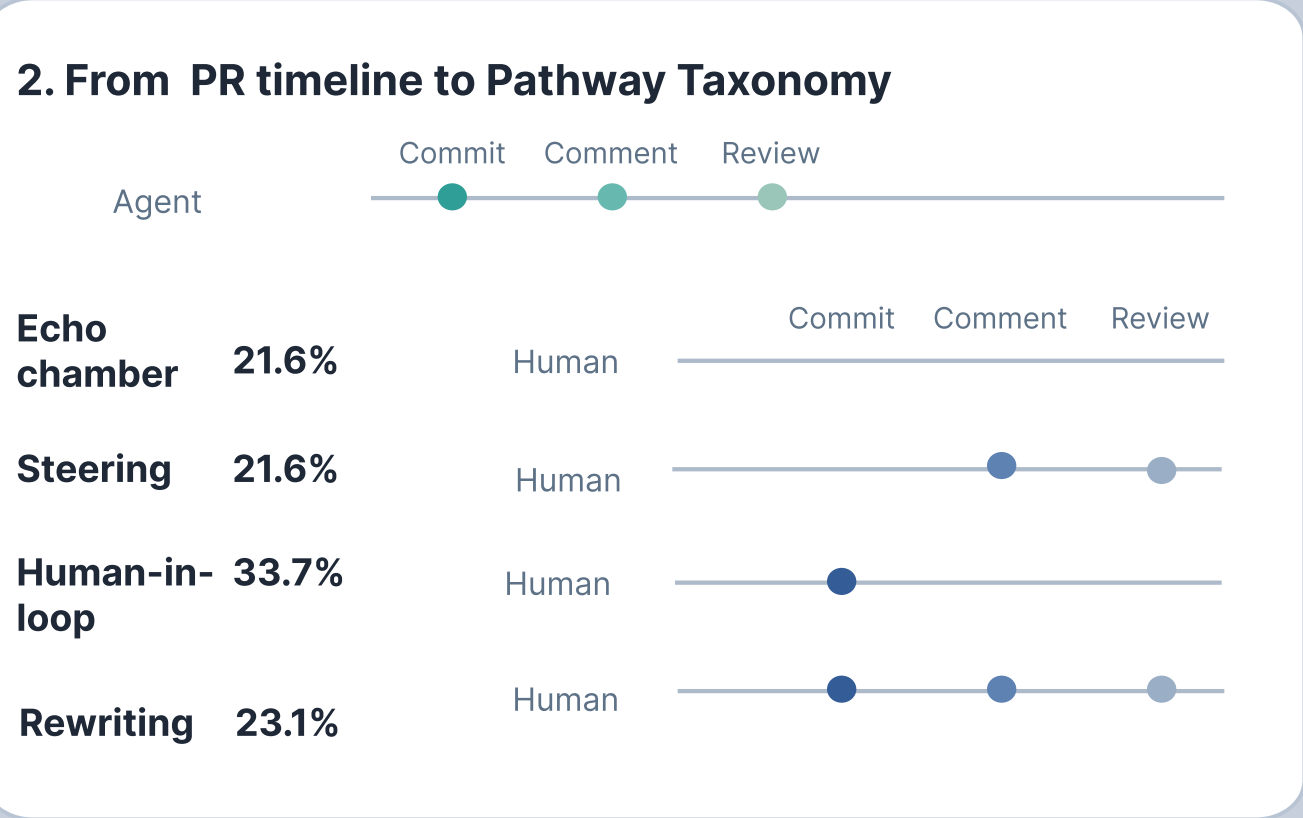
56.8%

With comments or Review

44.7%

Echo chamber

21.6%



Main findings

A. Fast human entry, often through comments

Agent PRs are more often comment-first and reach earlier median human response.

Comment-first

Median first response time

Agent

Human

55.5%

36.0%

0.03h

0.22h

B. Early attention does not mean hands-off integration

Merge rate without non-author commits

Agent

Human

Bot

11.98%

27.92%

46.79%

C. Who enters differs by handling mode

Earliest visible entrants are not uniform across pathways.

Human-in-loop

Steering

Rewriting

Novice

Mid

Expert

50.5%

25.9%

23.6%

49.2%

23.2%

27.5%

10.8%

43.3%

45.9%

Experts appear more often in steering and rewriting, while Novice/Mid are more common in human-in-loop.

